

Article

A Study of Direction-of-Arrival Estimation with an Improved Monopulse Ratio Curve Using Beamforming for an Active Phased Array Antenna System

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Abstract: When constructing a wireless communication network, the line of sight of radio waves is limited by the terrain features in a ground communication network. Also, satellite communication networks face capacity limitations and are vulnerable to jamming. Aviation communication networks can solve the above-mentioned problems. To construct seamless aviation communication networks, fast counterpart location estimation and efficient beam steering performance are essential. Among various techniques used for searching the counterpart's location, the monopulse technique has the advantage of quickly estimating the location through a simplified procedure. However, the nonlinear characteristics of the monopulse ratio curve, which are inevitably caused by the general antenna beam shape, both limit the location estimation range and reduce the estimated location accuracy. To overcome these limitations, a method that improves the estimation accuracy and extends the range by correcting the sum and difference patterns using the beamforming technique of active phased array antennas was proposed. An antenna system model suitable for aviation communication networks was presented, and the proposed model was experimentally proven to be effective. An average angle error of 0.021° was observed in the estimation of the accuracy of the antenna location.



Citation: Jung, J.; Satriyotomo, B.; Pyo, S. A Study of Direction-of-Arrival Estimation with an Improved Monopulse Ratio Curve Using Beamforming for an Active Phased Array Antenna System. *Electronics* **2023**, *12*, 4906. <https://doi.org/10.3390/electronics12244906>

Academic Editor: Jung-Dong Park

Received: 27 October 2023

Revised: 29 November 2023

Accepted: 4 December 2023

Published: 6 December 2023

Keywords: direction-of-arrival; beamforming; active phased array; antenna; monopulse

1. Introduction

A military communication network establishes a wireless communication network to support the rapid deployment of combat forces in emergency situations [1]. During the construction of the communication network, ground communication networks face challenges caused by the obstruction of radio visibility by terrain and man-made structures. For satellite communication networks, although they offer mobility, their data transfer capacity is limited relative to the associated costs, and they are constantly exposed to radio interference. Consequently, there is growing interest in establishing an aviation communication network utilizing aircraft [2].

As aircraft are in constant operation and motion, the antenna system crucial for effectively establishing an aviation communication network must possess rapid beam-steering performance. Additionally, it should be capable of steering the beam in a forward direction based on the azimuth angle criteria. The active phased array antenna system is an antenna system that electronically steers and shapes the beam by adjusting the amplitude and phase of the excited signals from each arrayed radiating element [2–8]. This allows rapid beam-steering and beam-shaping performance. Additionally, depending on the configuration of the radiating element array, this method enables forward beam steering based on an azimuth-based approach [9–11]. Therefore, we can conclude that an active phased array antenna system is suitable for establishing aviation communication networks.



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When in flight, an aircraft is constantly in a state of motion, so its location cannot be determined. To establish an aviation communication network, it is necessary to quickly and accurately detect the location of the communication counterpart.

Representative methods of detecting the location of a communication counterpart based on an active phased array antenna include ESPRIT [12–14], MUSIC [15–17], heuristic algorithms [18–20], and the monopulse technique [21–23]. ESPRIT estimates the precise direction of a signal using multiple antenna arrays. It is based on the principle that signals exhibit unique rotational invariances between different antennas in an array. Thus, ESPRIT assumes that a signal is invariant to the rotation between antennas and utilizes this assumption, along with the relative spacing between antennas, to accurately estimate the direction. This function makes ESPRIT particularly adept at accurately estimating signals in high-density antenna arrays; thus, it is known for its high-resolution nature. However, a disadvantage of ESPRIT is its slow estimated angle calculation due to the large number of eigenvalue decompositions performed [12]. The MUSIC technique uses signals received from a multiple antenna array to generate a spatial spectrum matrix. Through the Singular Value Decomposition (SVD) of the spatial spectrum matrix into eigenvectors and eigenvalues, the MUSIC algorithm calculates its estimated angles. In particular, the MUSIC technique has robust performance in terms of accurately estimating signals, especially in high-density antenna arrays. However, it requires significant computation, such as through the decomposition of training values and the detection of spectra and eigenvalues, resulting in a slow estimated angle calculation [16]. Heuristic algorithms are used to solve complex problems, helping researchers to identify approximate solutions quickly without seeking the optimal solution, as they rely on empirical knowledge. These algorithms typically operate efficiently by following heuristic rules associated with the characteristics of the problem in question. When applying a heuristic algorithm to a specific beamforming problem, the location of the counterpart is estimated by searching for the optimal phase weighting set. However, the estimated location calculation is also slow due to the number of cases considered, which exponentially increases in line with the number of arranged radiation elements [24].

In aviation communication networks, the calculation time required to estimate the location of the counterpart aircraft is very important. The monopulse technique calculates the estimated location based on the monopulse ratio curve, which is the synthesis of the sum and difference patterns formed via the monopulse comparator or excited signal weights. Not only can the sum and difference patterns be generated through simple signal synthesis, but based on the monopulse ratio curve, the location can also be estimated using a simple equation. Therefore, this method takes little time to estimate the location of the communication counterpart. However, given the beam shape of a general array antenna, the linear region of a monopulse ratio curve is limited. This phenomenon not only limits the location estimation range, but also increases the location estimation error. In addition, it increases the number of beams that must be formed for the location estimation to occur. To solve the above-mentioned problems, techniques such as the optimization of the design element values of the monopulse receiver and artificially expanding the beam width of the antenna have been studied. However, one disadvantage of using the monopulse receiver design element optimization technique is that it requires that a separate monopulse receiver is installed in addition to the active-phased array antenna system. Artificial antenna beamwidth expansion is the most representative method of expanding the linear region. However, its main disadvantage is that it cannot ideally solve the problem of linear region limitations.

In this paper, we studied a technique for the fast and accurate estimation of a communication counterpart's location based on the monopulse technique in an aircraft equipped with an active phased array antenna system to build an aviation communication network. A configuration of an aircraft-mounted antenna system suitable for establishing an aviation communication network was presented. Moreover, based on the presented antenna system, we studied a method for improving the linear characteristics of the monopulse ratio curve and more accurately estimating the location of a communication counterpart. The details of

the active phased array antenna system configuration for an aviation communication network, the monopulse ratio curve improvement method, and the performance verification of the proposed method through simulation are described in the second chapter.

2. Antenna System Configuration

This section describes the design of an antenna system suitable for both creating an aerial communication network and simulating the techniques presented in this paper.

2.1. Antenna System Configuration

For an antenna mounted on manned or unmanned airborne vehicles as part of an aerial communication network, beam formation and steering are necessary in all directions based on azimuth. To achieve this goal, the radiating elements must be adaptively arranged in a cylindrical shape based on the azimuth. However, the cylindrical-shaped adaptive array structure has many disadvantages with regard to radiation pattern prediction, RF (Radio Frequency) circuit configuration, antenna system mounting fixture manufacturing, and antenna system maintenance. To solve this problem, this study presented an array of polygonal shapes. Moreover, one side of the polygon was equipped with a planar array antenna (hereinafter referred to as a tile).

When constructing an antenna system with a polygonal tile array, beam steering must be performed using a single tile in response to the polygonal shape to carry out omnidirectional beam steering based on the azimuth. During a beam-steering operation based on an active phased array antenna, antenna gain deviation occurs as a result of the radiation characteristics of the radiating element and the influence of beam steering. Also, gain deviation due to beam steering causes the quality of communication to deteriorate. In this paper, considering such deviations and antenna system implementation (maintenance, implementation, complexity, etc.), the antenna system structure for aviation communication network only was set to resemble a 10-tile cylindrical array. In terms of antenna system requirements for wireless communication, the Ku-band was denoted as the operating frequency band, and the antenna gain based on a single beam was set to 20 dBi or higher. To satisfy the required gain performance, in this paper, radiating elements arranged in an eight (based on azimuth angle) by four (based on elevation angle) pattern in a single tile were arranged via uniform spacing in a horizontal line. For the aviation communication network, we assumed that the operating altitudes of aircraft required for establishing the main communication network were similar. Accordingly, we assumed that the antenna system did not perform beam steering based on elevation angle. Therefore, only four radiating elements were arranged based on the elevation angle. Figure 1 shows a conceptual diagram of the antenna system according to the above-mentioned settings.

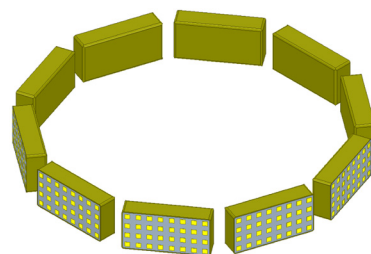


Figure 1. Proposed antenna system configuration.

2.2. Antenna System Detailed Settings and Radiation Characteristics Analysis

In this study, we assumed that the beams employed for establishing an aviation communication network are formed on a tile base. Based on the antenna system configuration presented in this paper, a total of 10 tiles were arranged. Therefore, single-tile reference beam steering must operate within a range from $+18^\circ$ to -18° based on the broad side of each tile. In a planar, linear, uniformly spaced array antenna, the distance between radiating elements is set so that only a single grating lobe exists in the visible region due to beam steering. The

distance between the radiating elements based on the maximum beam-steering angle required to satisfy these conditions is calculated through Equation (1), as follows:

$$d \leq \frac{(N-1)}{N} \frac{\lambda}{1 + \sin \theta_m} \quad (1)$$

where d is the distance between radiating elements, N is the number of radiating element arrays, λ is the wavelength, and θ_m is the maximum beam-steering angle.

For the azimuthal reference, eight elements are arranged and the maximum beam-steering angle is 18° . Meanwhile, for the elevation reference, four elements are arranged, and the maximum beam steering angle is 0° . Accordingly, the distance between each radiating element is set to 0.66λ and 0.75λ for azimuth and elevation, respectively.

Figure 2 shows the radiation characteristics based on a single tile according to the above conditions. Here, the single radiating element is assumed to be a microstrip patch, and the radiation characteristics of the single radiating element are applied.

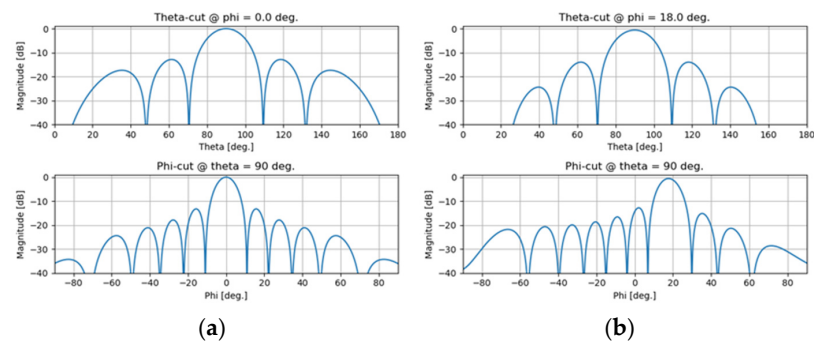


Figure 2. Radiation characteristics at (a) beam-steering angle with $\phi = 0^\circ$ and (b) beam-steering angle with $\phi = 18^\circ$.

As a result of the analysis, for the case in which there is no beam steering ($\phi = 0^\circ$), the half-power beamwidth (HPBW) based on the azimuth and elevation were 9.6° and 17.4° , respectively. Moreover, the gain was calculated to be about 21.01 dBi. When maximum beam steering ($\phi = 18^\circ$) was performed based on a single tile, the HPBW based on the azimuth and elevation were 10.2° and 17.4° , respectively, and the gain was about 20.14 dBi. As a result, the required gain of 20 dBi can be satisfied according to the antenna system configuration presented in this paper. In addition, the gain deviation due to single-tile-based beam steering for the azimuth-based one is less than 1 dBi.

2.3. Antenna System Geometry

In this paper, the Ku-band was selected as the frequency band for establishing an aviation communication network. Moreover, the frequency for antenna design was set to 15 GHz. Figure 3 shows the corresponding antenna system geometry.

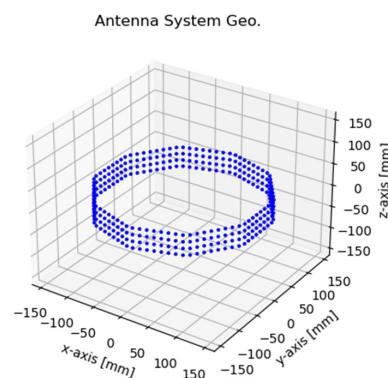


Figure 3. Proposed antenna system geometry.

The diameter of the antenna system according to the 10-tile cylinder arrangement is approximately 282 mm, and the height is approximately 45 mm.

Figure 4 shows the radiation characteristics of each tile in the antenna system when sequentially operated. In Figure, #1, #2, $\dots$, #10 represent tile indices. The detailed radiation characteristics of each tile are the same as the results in Figure 2a.

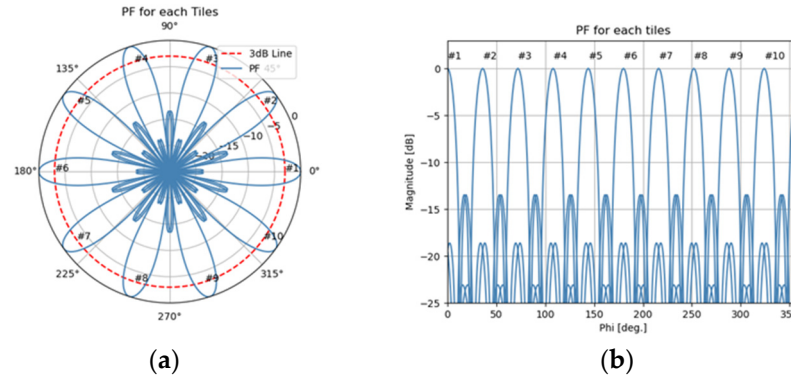


Figure 4. Radiation characteristics for each tile in (a) polar plot and (b) rectangular plot.

3. Monopulse Ratio Curve Improvement

In this section, we describe the monopulse ratio curve calculation method, the monopulse ratio curve calculation based on the proposed tile configuration, the general monopulse ratio curve improvement method (beam width expansion), and the proposed monopulse rate curve improvement method.

3.1. Monopulse Ratio Curve Calculation Method

The monopulse technique forms a sum pattern and a difference pattern based on the beam-forming technique and estimates the location of the communication partner based on the ratio of the signals received by them. Here, the ratio of the sum pattern and the difference pattern is called the monopulse ratio curve, and it is used as an indicator for estimating the communication counterpart's location.

Typically, the patterns for sum and difference signals are created using a monopulse comparator. However, in the case of an active phased array antenna, a pattern for the sum and difference signals can be formed through a phase-weighted combination of the excited signal of each of the arrayed radiating elements. Equation (2) shows the formula for the sum and difference patterns of the active phased array antenna based on the phase weighting combination, as follows:

$$P_{\Sigma} = S(\theta, \phi) \sum_n \sum_m \exp(jk\bar{r} \cdot \hat{r}_{(n,m)}) \quad (2)$$

$$P_{\Delta} = S(\theta, \phi) \sum_n \sum_m \exp(jk\bar{r} \cdot \hat{r}_{(n,m)} + jD_{(n,m)}) \quad (3)$$

where $S(\theta, \phi)$ is the radiation characteristic of a single radiating element (microstrip patch), n is the radiating element index in elevation reference (four elements), m is the radiating element index in azimuth reference (eight elements), k is the propagation constant, $\bar{r}$ is the unit vector in the r direction in a spherical coordinate system, $\hat{r}$ is the position vector of the (n, m) -th radiating element, and D is the phase weight for the difference pattern synthesis.

As can be seen in Equation (2), the sum pattern is a pattern without a phase shift weight, it is identical to the radiation characteristics of a single tile without beam steering. The phase weighting D for forming the difference pattern is set as shown in Equation (4):

$$D_{(n,m)} = \begin{cases} 0, & (1 \leq m \leq 4, \forall n) \\ \pi, & (5 \leq m \leq 8, \forall n) \end{cases} \quad (4)$$

Based on the sum and difference patterns, the monopulse ratio curve for estimating the counterpart's location is calculated as shown in Equations (5) and (6). Here, Equation (5) is the formula for the estimated angle, and Equation (6) is the formula for distinguishing the direction (left/right) of the communication counterpart relative to the direction of the broadside for tile, as follows:

$$an = \left| \frac{\Delta}{\Sigma} \right| \quad (5)$$

$$dir = \tan^{-1} \left(-Im \left(\frac{\Delta}{\Sigma} \right) \right) \quad (6)$$

where Σ is the sum signal, Δ is the difference signal, and $Im()$ is the imaginary output function.

The monopulse ratio curve is generally found to have a slope of '1', which provides a 1:1 correspondence between the calculated value of Equations (5) and (6) and the estimated angle. For this purpose, the coefficient (c_r) for monopulse ratio curve control is applied to Equation (5). The optimal monopulse ratio curve control coefficient (c_r) can be calculated using MMSE (Minimum Mean Square Error), as shown in Equation (7):

$$c_r^{opt} = \min_{c_r} \mathbb{E} |\phi - c_r an|^2, \quad (0 \leq \phi \leq \phi_c) \quad (7)$$

where ϕ_c is the range required for coefficient optimization.

The estimated angular accuracy of the monopulse ratio curve with the calculated coefficients is given by the mean absolute error (MAE), as shown in Equation (8).

$$MAE = |\phi - c_r^{opt} an|, \quad (0 \leq \phi \leq \phi_c) \quad (8)$$

3.2. Monopulse Ratio Curve for the Presented Tile

In this section, the monopulse ratio curve for a single tile was analyzed based on the antenna of the system configuration discussed in Section 2.

Figure 5 shows the sum and difference patterns of Equations (5) and (6) according to the presented tile configuration. The HPBW of the sum pattern is 9.6° .

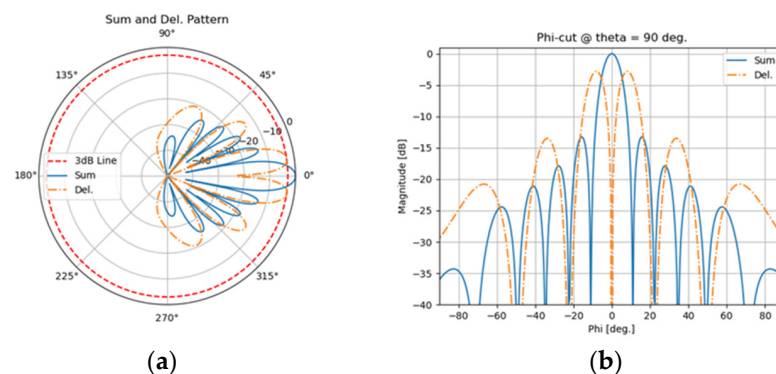


Figure 5. Single-tile radiation characteristics in (a) polar plot and (b) rectangular plot.

Table 1 shows the optimum c_r and the location estimation accuracy, which are calculated based on Equations (7) and (8) for the presented tile.

Table 1. For presented tile, estimation accuracy for monopulse ratio curves according to ϕ_c .

ϕ_c	3°	3.6°	4.2°	4.8°	5.4°	6.0°
c_r^{opt}	6.634	6.534	6.418	6.126	5.948	5.746
MAE mean	0.026	0.044	0.065	0.129	0.180	0.244
MAE maximum	0.075	0.123	0.192	0.404	0.562	0.769

As the estimation range of the communication counterpart's locations expands, it becomes difficult to optimize the linear characteristics of the monopulse ratio curve; thus, the location estimation accuracy decreases. Figure 6 shows the monopulse ratio curves for $\phi_c = 3.6^\circ$ and $\phi_c = 6.0^\circ$.

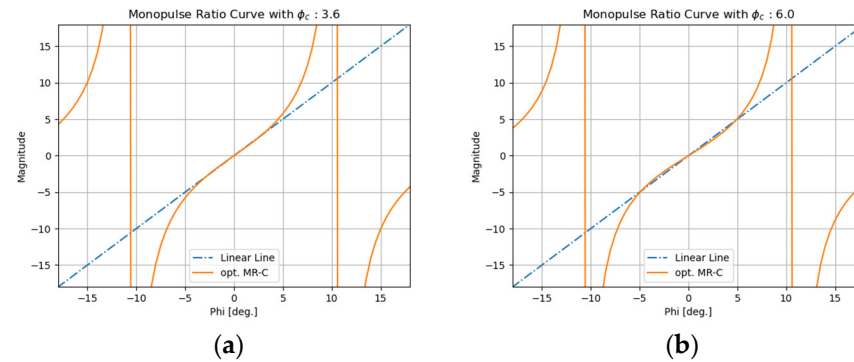


Figure 6. Monopulse ratio curves in (a) $\phi_c = 3.6^\circ$ and (b) $\phi_c = 6.0^\circ$.

3.3. General Monopulse Rate Curve Improvement Method Based on the Presented Tile

The most representative method for improving the linear characteristics of the monopulse ratio curve is by expanding the beam width of the antenna. In the presented antenna system, the method employed to expand the beam width of a single-tile antenna involves disabling some of the radiating elements within a single tile. In this paper, for the radiating elements within a single tile, the beam width was expanded for the cases that involved using either six or four azimuth-based radiating elements. Figure 7 shows the status of the radiating element operating configuration used to expand the beam width based on the above method.

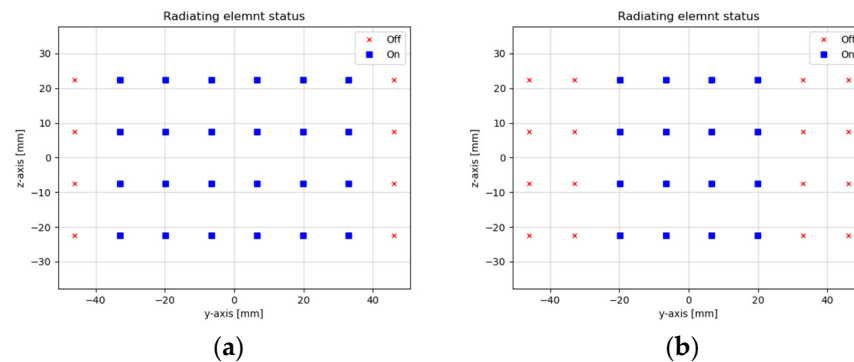


Figure 7. Radiating element status with (a) six azimuth-based elements and (b) four azimuth-based elements.

Figure 8 shows the radiation characteristics (sum and difference patterns) according to each operating state. The HPBW of the sum pattern for six azimuth-based radiating elements is 13.0° , while the figure for four azimuth-based radiating elements is 19.8° .

Table 2 shows the optimum c_r and the location estimation accuracy, which are calculated based on Equations (7) and (8) for each radiating element's operating state.

Table 2. For general method, estimation accuracy for monopulse ratio curves according to ϕ_c .

		ϕ_c	4.0°	4.5°	5.0°	5.5°	6.0°	6.5°
6 elements	c_r^{opt}		8.854	8.756	8.648	8.524	8.388	8.236
	MAE mean		0.035	0.049	0.069	0.094	0.124	0.16
	MAE max		0.103	0.148	0.209	0.284	0.379	0.496

Table 2. Cont.

	ϕ_c	4.0°	5.0°	6.0°	7.0°	8.0°	8.5°
4 elements	c_r^{opt}	13.586	13.456	13.294	2013.1.2	12.874	12.612
	MAE mean	0.015	0.029	0.051	0.08	0.124	0.179
	MAE max	0.043	0.087	0.153	0.249	0.382	0.562

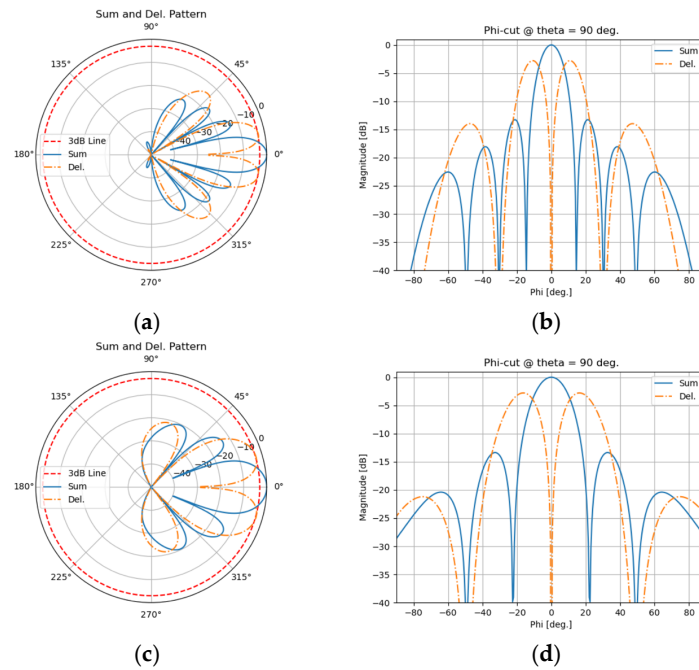


Figure 8. Sum and difference patterns in (a) six elements with polar plot and (b) rectangular plot and (c) four elements with polar plot and (d) rectangular plot.

Figure 9 shows the monopulse ratio curves for six azimuth-based radiating elements for which $\phi_c = 4.5^\circ$ and $\phi_c = 6.5^\circ$, as well as four azimuth-based radiating elements for which $\phi_c = 6.0^\circ$ and $\phi_c = 8.0^\circ$.

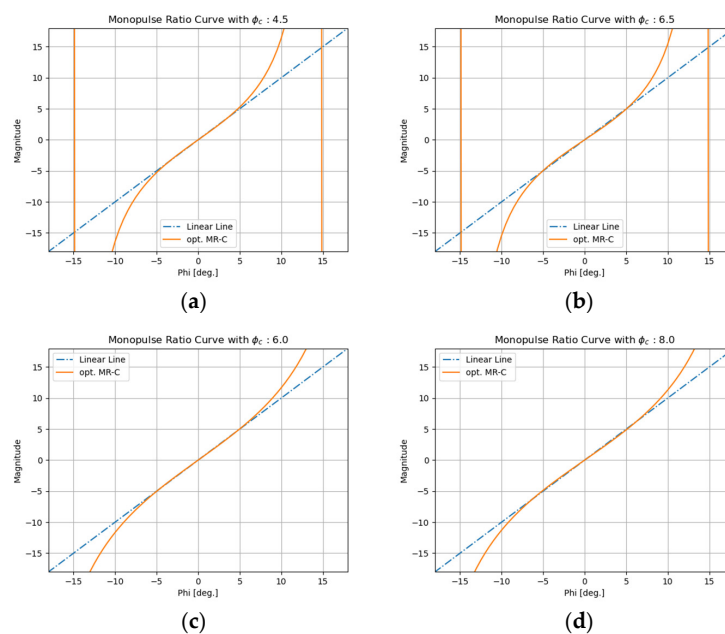


Figure 9. Monopulse ratio curves in (a) $\phi_c = 4.5^\circ$, (b) $\phi_c = 6.5^\circ$, (c) $\phi_c = 6.0^\circ$, and (d) $\phi_c = 8.0^\circ$.

The results show that the linear region of the monopulse ratio curve expands as the beam width expands. Moreover, as shown in Figure 6, when the location estimation range is expanded, the accuracy of the estimation of the location decreases.

3.4. Proposed Monopulse Rate Curve Improvement Method Based on the Presented Tile

An active phased array antenna can shape the beam into a specific shape through the amplitude and phase weighting of the excited signal. In this paper, the beamforming technique was used to improve the monopulse ratio curve. Thus, the sum and difference patterns were synthesized to expand the linear region of the monopulse ratio curve.

The array factor of the active phased array antenna with the weighted amplitude and phase of the excited signal is given in Equation (9), as follows:

$$AF = \sum_n \sum_m W_{(n,m)}^a \exp(jk\vec{r} \cdot \hat{r}_{(n,m)} + jW_{(n,m)}^p) \quad (9)$$

where $W_{(n,m)}^a$ and $W_{(n,m)}^p$ are the amplitude and phase weighting of the (n, m) -th radiator, respectively.

The sum and difference pattern for calculating the monopulse ratio curve based on the AF of a single tile is shown in Equations (10) and (11).

$$P_\Sigma = S(\theta, \phi) \sum_n \sum_m W_{(n,m)}^a \exp(jk\vec{r} \cdot \hat{r}_{(n,m)} + jW_{(n,m)}^p) \quad (10)$$

$$P_\Delta = S(\theta, \phi) \sum_n \sum_m W_{(n,m)}^a \exp(jk\vec{r} \cdot \hat{r}_{(n,m)} + jW_{(n,m)}^p + jD_{(n,m)}) \quad (11)$$

Based on Equations (10) and (11), the optimum W^a and W^p should be obtained to stably extend the linear section of the monopulse ratio curve. Thus, a type of heuristic algorithm was used, namely the vegetative propagation by runner (VPR) optimization algorithm [25].

The VPR algorithm is an algorithm that mimics the ecology of plants that reproduce through vegetative organs, and it improves a local optima problem by including the aging phenomena of plants and the continuous surrounding reproduction process. The VPR algorithm identifies the best cost evaluation location by moving and propagating based on the soil quality evaluation value (P_b , S_b) obtained through the tap roots and lateral roots formed by each plant, as shown in Equation (12), as follows:

$$Run^i = c_s rand() (S_b^i - X^i) + c_p rand() (P_b - X^i) \quad (12)$$

where i represents the index of the i -th plant, Run represents the position of the vegetative organs, c_p and c_s are the vegetative organ generation coefficient based on the tap root and lateral root, respectively, $rand()$ represents the uniformly distributed random numbers, and X^i represents the position of the i -th plant.

The position of the next-generation plant is calculated by accumulating the positions of the vegetative organ and the previous generation plant, as shown in Equation (13), as follows:

$$X^i[t+1] = X^i[t] + run^i[t+1] \quad (13)$$

where t is an index representing the generation that corresponds to the number of algorithm iterations.

The next-generation plant's tap root is formed at the propagated location on the creeping stem. Therefore, the location of the tap root used for evaluating soil quality is the same as the location of the plant. Lateral roots then form around the tap roots. Therefore, the location of the entourage is expressed in Equation (14), as follows:

$$lat^i = X^i + c_z randn() \quad (14)$$

where c_z is the entourage occurrence coefficient, and $randn()$ is a normally distributed random number.

The VPR algorithm includes plant aging procedures. This procedure simulates the process through which plants absorb fewer nutrients as they age, and it is equivalent to Equation (15).

$$P_b[t + 1] = P_b[t] \exp(c_t \text{ age}) \quad (15)$$

where c_t is the aging coefficient, and age is the plant aging time, which increases by '1' according to the number of the algorithm iteration. Finally, age is initialized to '0' when a new, highest-quality location is identified as plants propagate.

In this paper, the values corresponding to the plant location in the VPR algorithm are amplitude and phase weighting. For the antenna system presented in this paper, the estimation of the location is not performed in the elevation angle direction based on assumptions. Therefore, a is unrelated to b , and each weight can be expressed as a vector, as shown in Equations (16) and (17).

$$W^a = [W_0^a, W_1^a, \dots, W_7^a] \quad (16)$$

$$W^p = [W_0^p, W_1^p, \dots, W_7^p] \quad (17)$$

The cost function for applying the VPR algorithm was defined as shown Equation (18).

$$\text{cost} = 1000 \text{std}(|\frac{P_\Delta}{P_\Sigma}(\phi)|) + \frac{1}{\max(P_\Sigma)}, \quad (0 \leq \phi \leq \phi_o) \quad (18)$$

where $\text{std}()$ is the deviation calculation function, and 1000 is the deviation evaluation weighting factor coefficient for weighting the evaluation of the deviation.

In this study, the VPR algorithm was set to search for the minimum cost evaluation value. Therefore, a weight with a small deviation in the monopulse ratio curve according to ϕ was searched. In addition, when calculating the cost function, the reciprocal of the maximum value of the sum signal was added. We took this step to induce the linear characteristics of the monopulse ratio curve and, at the same time, set the synthesized sum pattern to achieve maximum gain. The deviation weighting coefficient was applied to increase the proportion of the monopulse ratio curve compared to the sum pattern gain for calculating the cost function.

Table 3 shows the parameters of the VPR algorithm for searching for optimal weighting values. The VPR algorithm parameters for amplitude weight search and phase weight search were each set in a different manner because the conditions for each weight search were different.

Table 3. VPR algorithm main variable settings.

Parameter	W^a	W^p
c_s	0.2	0.6
c_p	0.4	0.9
c_z	0.05	0.147
c_t	0.004	

Figure 10 shows the cost evaluation value based on the iteration of the VPR algorithm and the optimal weighting, as searched for using the VPR algorithm.

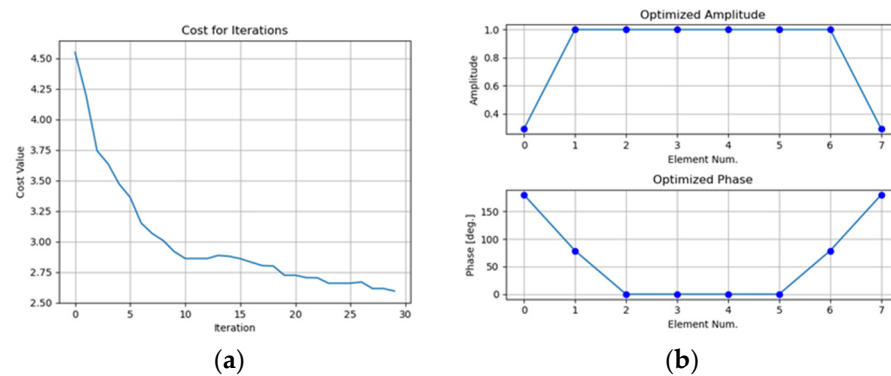


Figure 10. VPR algorithm results: (a) best cost value according to algorithm iteration; (b) optimized amplitude and phase weighting.

The algorithm stop criterion was set as the cost with 2.61. The optimal amplitude and phase weighting processes were found to be Equations (19) and (20), as shown in Figure 8b.

$$W^a = [0.29, 1, 1, 1, 1, 1, 1, 0.29] \quad (19)$$

$$W^p = [180, 78.75, 0, 0, 0, 0, 78.75, 180] \quad (20)$$

Figure 11 shows the radiating characteristics of the application of the optimized weights to Equations (10) and (11).

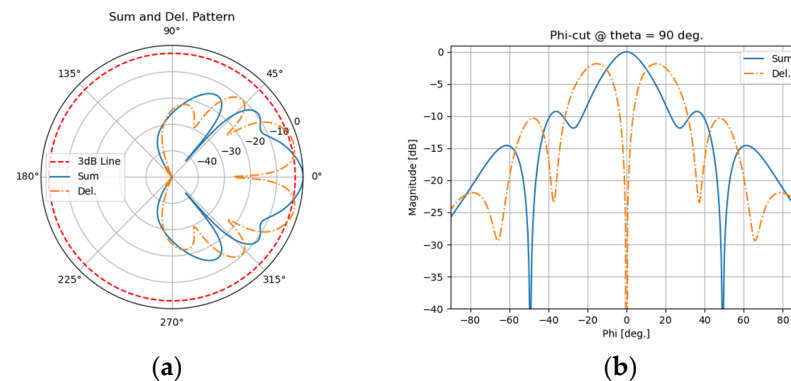


Figure 11. Single-tile-optimized radiation characteristics in (a) polar plot and (b) rectangular plot.

Based on $\phi_c = 18^\circ$ and the sum and difference patterns shown in Figure 11, using Equation (7), c_r was found to be 10.268.

Figure 12 shows the monopulse ratio curve with the sum and difference patterns in Figure 9 and $c_r = 10.268$. As a result of this analysis, the MAE mean was 0.01, and the MAE maximum was 0.04. Compared to the general monopulse ratio curve improvement method, the searching range and location estimation error for the communication counterpart have been improved.

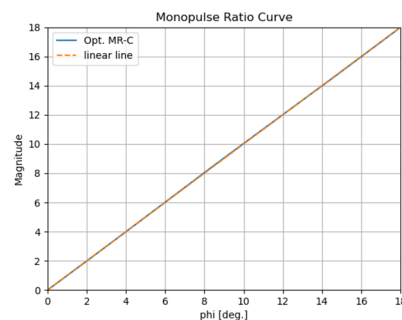


Figure 12. Improved monopulse ratio curve.

When estimating the location of a communication counterpart using the monopulse technique, the received signal strength cannot be confirmed because the monopulse ratio curve is the ratio of the sum and difference signals. Therefore, when estimating the location, the received signal strength based on the sum pattern must also be checked. As a result of the sum pattern analysis shown in Figure 11, the gain difference within the counterpart's location estimation region is approximately 6.9 dB. As the gain difference decreases, the ambiguity involved in estimating the location of the counterpart based on the received signal strength using the sum pattern decreases. In this study, to solve this problem, the received signal was synthesized using a sum and difference pattern to maintain, in the location estimation region, a relatively uniform received signal strength. The synthesized received signal strength is shown in Equation (21).

$$PF_{RSSI} = P_{\Sigma} + 0.4195P_{\Delta} \quad (21)$$

Figure 13 shows the result of converting the synthesized received signal strength into a radiation pattern. In Figure, #1, #2, $\dots$, #10 represent tile indices.

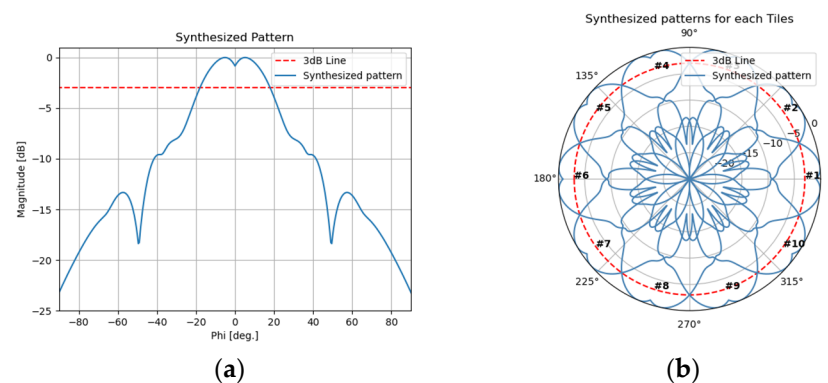


Figure 13. Synthesized patterns in (a) single tile and (b) antenna system.

Table 4 shows the results generated by comparing the estimation performances of the monopulse ratio curves of the proposed and other techniques (eight elements (presented tile), six elements, and four elements). Here, the monopulse rate curve was chosen for comparison with the proposed technique, considering a case in which the MAE mean was approximately 0.05.

Table 4. Estimation accuracy comparison.

Case	Presented Tile (8 Elements)	General Method		Proposed Method
		6 Elements	4 Elements	
ϕ_c	3.6°	4.5°	6.0°	18.0°
MAE mean	0.044	0.049	0.051	0.01
Beamforming num.	5	4	3	1

The “beamforming number” mentioned in Table 4 represents the beamforming number at 36° , which is the divide region for a single tile, divided into the linear region of the selected monopulse ratio curve. Figure 14 shows the beamforming results based on the normalized pattern.

As a result of the comparative analysis, we can confirm that the performance of the method proposed in this paper was improved for its use in aviation communication networks.

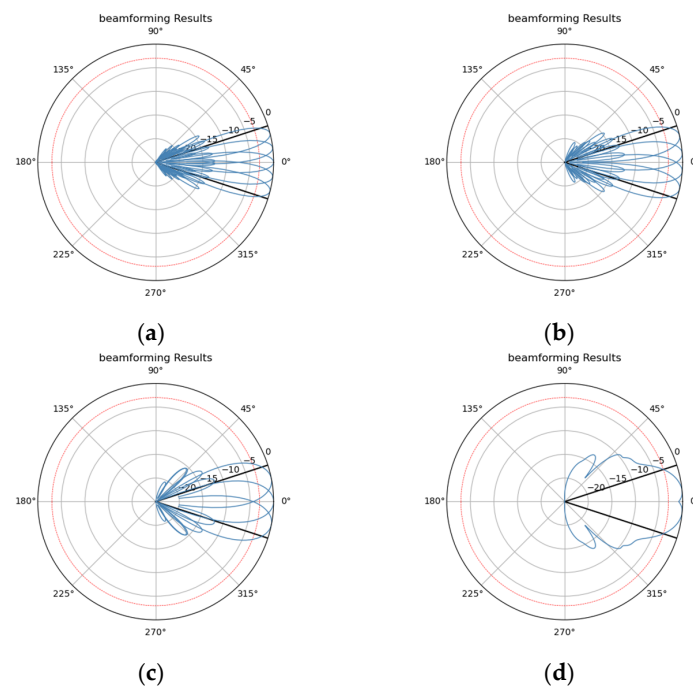


Figure 14. Beamforming results in (a) the presented tile (8 elements), with (b) 6 elements, (c) 4 elements, and (d) with the proposed method.

4. Simulation

In Section 4, the performance of the communication counterpart location estimation method was confirmed through simulations based on the results of the method used for the improvement of the monopulse ratio curve. As a basic assumption for the simulation experiment, we set the elevation angle at the location of the counterpart to confirm that the aviation communication network is randomly distributed within a range of $\pm 3^\circ$. This occurs due to the fact that even if the aircraft is similar, the operating altitude may be somewhat different.

4.1. Single Tile

Based on the proposed technique, a single-tile-based simulation experiment was performed to analyze the accuracy of the estimation of the counterpart location using a monopulse ratio curve with an improved linear section.

Figure 15 shows the counterpart's location (θ, ϕ), as set using a random function, along with the improved sum and difference patterns. The counterpart location based on the azimuth (ϕ) is 12.702° , whereas based on the elevation, (θ) is 91.711° . As a result of the simulation, the azimuth-based counterpart's location, as confirmed using the proposed technique, was confirmed to be 12.689° . Accordingly, the azimuth-based location estimation error was found to be approximately 0.013° .

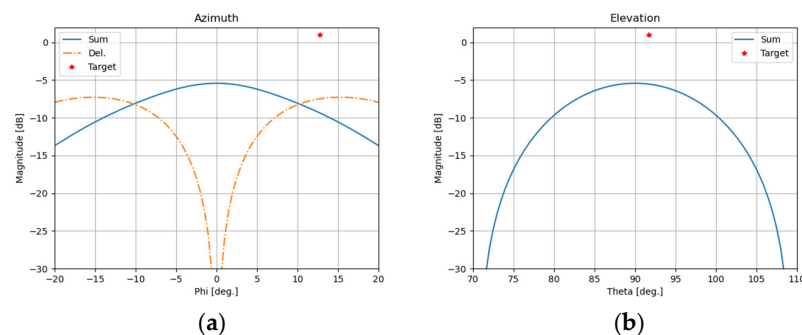


Figure 15. Sum and difference patterns and counterpart locations in (a) azimuth and (b) elevation.

To confirm the performance of the proposed technique, 100 simulation experiments were conducted for the estimation of the counterpart location based on a single tile. Figure 16 shows the counterpart's location determined based on random estimation, along with the azimuth-based estimation error estimated via the proposed method.

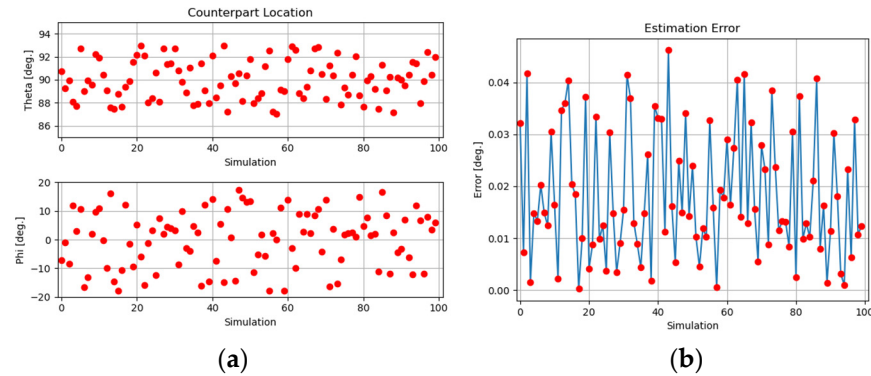


Figure 16. Simulation results for single tile: (a) counterpart location; (b) estimation error.

The analysis results show that for 100 simulations, the average and maximum azimuth-based estimation errors are 0.019° and 0.046° , respectively. The analysis results show that the counterpart's location estimation can be smoothly performed using the presented technique. For the analyzed data, the average azimuth-based estimation error is 0.046° , which is larger than the MAE mean shown in Table 4, as determined based on the optimized monopulse ratio curve. This result reflects the influence of the counterpart's location when distributed within $\pm 3^\circ$ based on the elevation.

4.2. Antenna System

In this paper, the presented antenna system is the 10-tile cylinder arrangement structure. Therefore, to estimate the counterpart's location, each tile must be sequentially operated, and the operating tile must be determined based on the synthesized received signal strength determined via Equation (21). Afterwards, the counterpart's location can be estimated based on the improved monopulse ratio curve of the determined tile. Based on the presented antenna, a system-based simulation experiment was performed to analyze the accuracy of the estimation of the counterpart location. Figure 17 shows the counterpart location (star marker) set using a random function and presented antenna system configuration. The counterpart location based on the azimuth (ϕ) is 198.450° , and the counterpart location based on the elevation (θ) is 87.7191° .

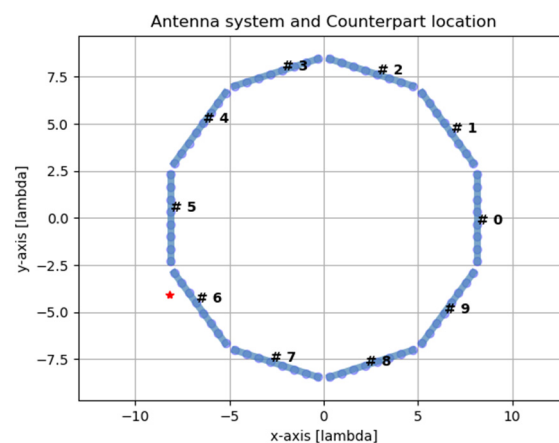


Figure 17. Antenna system configuration and counterpart location.

Figure 18 shows the received signal strength and the location estimation results based on sequential tile operation. As a result of the simulation, the received signal strength is

the highest in tile #6. The counterpart location can be calculated based on the location of tile #6 on the antenna system and the corresponding location estimate value. As a result of this analysis, the azimuth-based counterpart's location, as confirmed via the proposed technique, was found to be 198.478° . Accordingly, the azimuth-based location estimation error was confirmed to be approximately 0.028° .

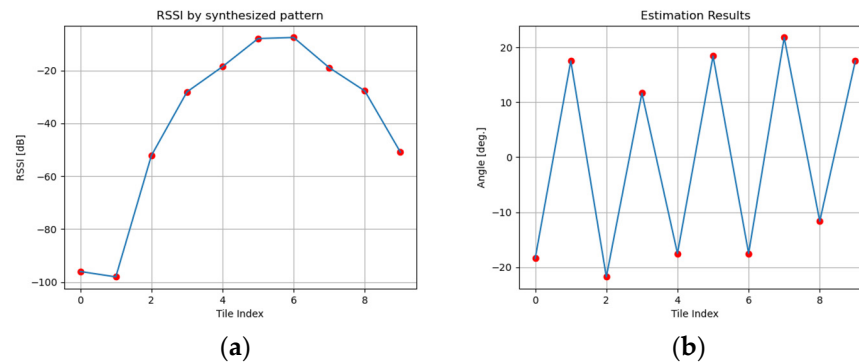


Figure 18. Simulation results: (a) RSSI via synthesized pattern; (b) estimation results.

To confirm the performance of the proposed technique, 100 simulation experiments were conducted for the estimation of the counterpart's location based on the presented antenna system.

Figure 19 shows the counterpart's location set based on random estimation, and it shows the azimuth-based estimation error estimated using the proposed method. The analysis results show that for 100 simulations, the average and maximum azimuth-based estimation errors are 0.021° and 0.042° , respectively.

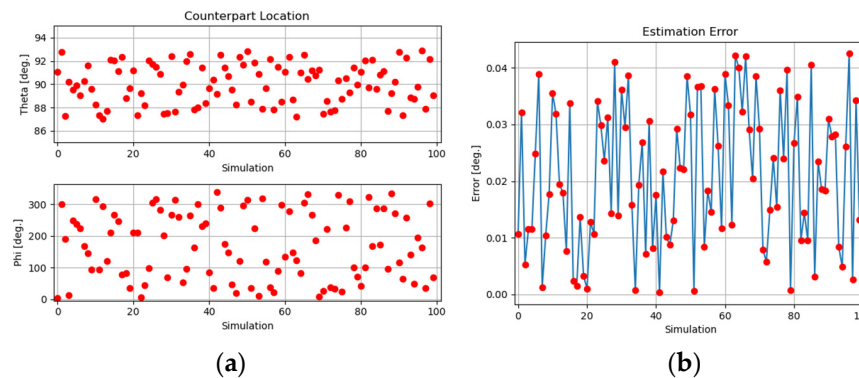


Figure 19. Simulation results for antenna system: (a) counterpart location; (b) estimation error.

5. Conclusions

In this paper, we investigated a counterpart estimation technique for establishing an aviation communication network. Considering that communication stations for the networks are airborne stations located in mobile and maneuvering environments, the estimation of fast antenna locations with fast beam-steering performance is required. Rapid beam-steering performance can be achieved by utilizing an active phased array antenna system. Among the various communication counterpart location estimation techniques available, the monopulse technique can produce the fastest estimation locations. However, due to the nonlinearity of the monopulse ratio curve, the estimation range is limited, and the estimation accuracy is low. To solve these problems, in this paper, we present a technique for improving the estimation accuracy while extending the estimation range by correcting the sum and difference patterns based on the beamforming technique using an active phased array antenna. To verify the performance of the proposed technique, we presented an antenna system configuration suitable for the construction of an airborne communication network. Also, based on the presented antenna system, the performance of the proposed technique was verified, and we

found that the error of the counterpart location estimation average error was 0.021° . We also believe that the technique proposed in this paper can be used to improve the performance of a tracking system via the monopulse technique.

Author Contributions: Conceptualization, S.P.; methodology, J.J.; validation, J.J. and B.S.; formal analysis, S.P. and J.J.; investigation, B.S.; data curation, B.S.; writing—original draft preparation, J.J. and B.S.; writing—review and editing, J.J.; supervision, S.P.; project administration, S.P.; funding acquisition, S.P. All authors have read and agreed to the published version of the manuscript.

Funding: This research was supported by the research fund of Hanbat National University in 2023 and this work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korean government (Ministry of Science and ICT) (NRF-2021R1A2C2011560).

Data Availability Statement: Data are contained within the article.

Conflicts of Interest: The authors declare no conflict of interest.

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